Lista Derivadas

L5.10

7) Max e Min

$f(x) = \frac{x}{1+x^2}$; $[2, 2]$ $f'(x) = \frac{(1)(1+x^2) - x(2x)}{(1+x^2)^2} = \frac{1-x^2}{(1+x^2)^2}$ //

0 $1 - x^2 = 0 \Rightarrow x = \pm 1$

Pontos

$f(-2) = \frac{-2}{1+4} = \frac{-2}{5}$ // $f(2) = \frac{2}{5}$

$\text{Min} = -1 : f(-1) = -0.5$ //

$\text{MAX} = 1 : f(1) = -0.5$ //

$f(-1) \frac{-1}{1+1} = \frac{-1}{2}$ // $f(1) = \frac{1}{2}$

8) g) $G - G(x)\ xe^x \rightarrow G'(x) = (x)' \cdot e^x + x \cdot (e^x)' = e^x + xe^x$

$e^x(1+x)$ $e^x(1+x)$

0 $G'(x) : e^x(1+x) : 0$

$e^x > 0 \rightarrow 1 + x : 0 \rightarrow x = -1$ //

Sinal: $x < -1 : G'(x) < 0$

Crescente Decrescente

$G(-1) = -1 \cdot e^{-1} = \frac{-1}{e}$ $x > -1 : G'(x) > 0$

$\nearrow (-1, +\infty)$ $\searrow (-\infty, -1)$ MAX R: $\nexists$ MIN R: $x = -1 \rightarrow -1/e$

15) c) $y = -1/4\, x^4 + 5/3\, x^3 - 2x^2$ $y' = -x^3 + 5x^2 - 4x$

Pontos

$y' = \dots = -x(x^2 - 5x + 4) = -(x-1)(x-4) = \text{raiz} = 0, 1, 4$

Crescente e Decrescente

$x < 0$ $y' < 0 = \searrow$; $0 < x < 1 : y' > 0 = \nearrow$; $1 < x < 4 : y' < 0 : \searrow$ $x > 4 : 6 > 0 : \nearrow$

0 MIN

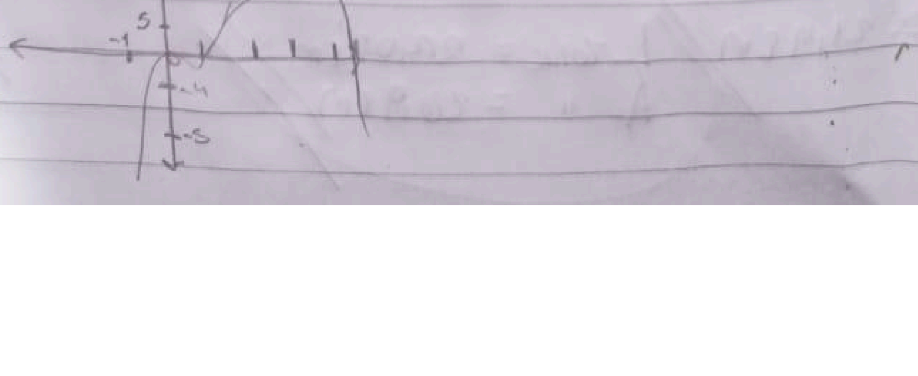
1 MAX

4 MAX

Concavidade

$y'' = -3x^2 + 10x - 4$

Gráfico



D) $Y = X + \frac{2}{X}$ $\quad Y' = 1 - \frac{2}{X^2}$

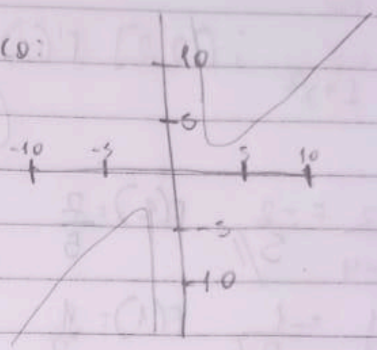
$1 - \frac{2}{X^2} = 0 = X^2 = 2 \Rightarrow X = \pm\sqrt{2}$

CRC E DECRE

↳ $|x| < \sqrt{2} : Y' < 0$ ↘

↳ $|x| > \sqrt{2} : Y' > 0$ ↗

MIN = $\sqrt{2}$ $\quad$ MAX = $-\sqrt{2}$ $\quad$ GRAFICO:



L5.12

7) A -> A' = 4KM ARROIO

A = (0, 4)

B = (12, 2)

RETA

$M = \frac{2-(-4)}{12-0} = \frac{6}{12} = \frac{1}{2}$

PONTO A'

$Y + 4 = 1/2\, x \rightarrow Y = 1/2\, x - 4$

$Y = 0 = 0 = \frac{1}{2}x - 4 = 8$

15) X = LARGURA E Y = ALTURA $\quad XY = 375\,cm^2$

⟷ $\quad$ ↑ $\quad$ ↳ $Y = \frac{375}{X}$

$X + 4,5$ $\quad$ $Y + 5,5$

AREA (I) = A

↳ $(X+4,5)(Y+5,5) = (X+4,5)\left(\frac{375}{X}+5,5\right) = \frac{1687,5}{X} + 5,5X + 399,75$

AREA(X) = A'

$-\frac{1687,5}{X^2} + 5,5 \rightarrow 0 \rightarrow \frac{1687,5}{X^2} = 5,5 = X^2 = \frac{1687,5}{5,5} = \sqrt{306,82}$

$\simeq 17,52\,cm$

$\frac{375}{X} = \frac{375}{17,52} \simeq 21,42\,cm$

L TOTAL = 2,2,02 cm

A $\quad$ " $\quad$ = 26,92 cm

26)

2m $\quad V = 6 \cdot L \cdot H = 2L \cdot H = 3 = L \cdot H = \frac{3}{2}$

$3m^3$

Altura $= 2(LH) + 2(LH) + 2(LL)$

$LH = \frac{3}{2} \quad H = \frac{3}{2L} \rightarrow A(L) = 2\left(\frac{3}{2}\right) + 4\left(\frac{3}{2L}\right) + 4L$

$\frac{3+6}{L} + 4 \rightarrow A'(L) = -\frac{6}{L} + 4 \rightarrow ① \rightarrow -\frac{6}{L^2} + 4 = 0 \quad L^2 = \frac{3}{2} \quad L =$

$\rightarrow \sqrt{3/2} = 1{,}225\,m \quad H = \frac{3}{2L} = \frac{3}{2\sqrt{3/2}} = \frac{3}{2}\sqrt{\frac{2}{3}} = \frac{\sqrt{6}}{2} = 1{,}225\,m$

Comprimento : 2 m

Altura : 1,225 m

Largura : ||

Ls.14

14) $\lim\limits_{x \to \infty^+} = x^2\left(e^{1/x} - 1\right) \quad T = 1/x \rightarrow 0^+$

$x^2 = \frac{1}{T^2}, \quad e^{1/e} - 1 = e^t - 1$

$\lim\limits_{x \to \infty^+} x^2\left(e^{1/x} - 1\right) = \frac{e^t - 1}{T^2} \Big\} \lim T \to 0^+$

$y_1' = \lim\limits_{t \to 0} \frac{e^t}{2T} \qquad y_2' \lim\limits_{T \to 0} \frac{e^t}{2} = \frac{1}{2}$

$\lim\limits_{x \to \infty^+} x^2\left(e^{1/x} - 1\right) = 1/2$

22) $\lim\limits_{x \to \infty} \frac{\ln x}{\sqrt[3]{x}} \rightarrow \frac{\infty}{\infty} \rightarrow$ L'Hopital $y' = \frac{d}{dy}(\ln x) = \frac{1}{x} \cdot \frac{d}{dx}\left(x^{1/3}\right) = \frac{1}{3}x^{-2/3}$

$\lim\limits_{x \to \infty} \frac{1/x}{1/3\,x^{-2/3}} = \frac{3x^{-1}}{x^{-2/3}} = 3x^{-1+2/3} = 3x^{-1/3} = 0$

$\lim x \to \infty$